

Filip Bělík

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PhD Candidate

Department of Mathematics and the Scientific Computing and Imaging Institute
University of Utah, Salt Lake City, UT

EDUCATION

• University of Utah

PhD in Mathematics: Applied & Computational Mathematics

Fall 2022 - Present

Salt Lake City, UT

◦ Advisors: [Dr. Akil Narayan](#) and [Dr. Christel Hohenegger](#)

◦ Expected graduation in May 2027

• Gustavus Adolphus College

BA in Mathematics with honors; BA in Computer Science

Fall 2018 - Spring 2022

St. Peter, MN

TEACHING EXPERIENCE

• Engineering Calculus II (Series and Multivariate and Vector Calculus)

Instructor of Record

Spring 2026

University of Utah

◦ Preparation of syllabus, lectures, assignments, quizzes, and exams

◦ Managed team of teaching assistants and helped develop lab materials

◦ Class size of 40

• Engineering Calculus I (Single Variable Calculus)

Instructor of Record

Spring 2025

University of Utah

◦ Preparation of syllabus, lectures, assignments, quizzes, and exams

◦ Managed team of teaching assistants and helped develop lab materials

◦ Class size of 38

• Graduate Applied Linear Operators and Spectral Methods

Teaching Assistant - Office hours and grading

Fall 2025

University of Utah

• Graduate Numerical Linear Algebra

Teaching Assistant - Grading

Spring 2024

University of Utah

• Mathematics of Medicine

Laboratory Instructor

Spring 2023

University of Utah

◦ Lectured and guided group work for MATLAB-based modeling assignments

PEER-REVIEWED PUBLICATIONS

1. H. Crandall, T. Schuessler, **F. Bělík**, et al. "Cuffless hemodynamic monitoring with physics-informed machine learning models". In: *Nature Communications* 17.1, 6429 (May 14, 2026). DOI: [10.1038/s41467-026-72693-1](https://doi.org/10.1038/s41467-026-72693-1). URL: <https://doi.org/10.1038/s41467-026-72693-1>

MANUSCRIPTS UNDER REVIEW

1. **F. Bělík**, A. Narayan, and J. Turnage. "Constructive Tchakaloff results and well-conditioned quadrature through randomized least squares". Under review at *Constructive Approximation*. 2026. arXiv: [2609.09506](https://arxiv.org/abs/2609.09506) [[math.NA](#)]
2. **F. Bělík**, Y. Chen, and A. Narayan. "Greedy Rational Approximation for Frequency-Domain Model Reduction of Parametric LTI Systems". Under review at *Research in the Mathematical Sciences*. 2025. arXiv: [2512.23814](https://arxiv.org/abs/2512.23814) [[math.NA](#)]
3. **F. Bělík**, J. Chan, and A. Narayan. "Efficient and Robust Carathéodory-Steinitz Pruning of Positive Discrete Measures". Under review at *BIT Numerical Mathematics*. 2025. arXiv: [2510.14916](https://arxiv.org/abs/2510.14916) [[math.NA](#)]

ACADEMIC SERVICE AND LEADERSHIP

- **Co-organizer: Conference Minisymposium**

November 2026

SIAM Conference on the Mathematics of Data Science

- Minisymposium title: Dimension Reduction and Data Selection for Model Order Reduction
- Corresponded with speakers and submitted information to the conference portal

- **President, University of Utah SIAM Student Chapter**

Fall 2024 - Spring 2026

University of Utah

- Led programming for undergraduate and graduate students, including mathematical modeling competitions, student-led conferences, professional development events, and social activities

- **Organizer, 2026 Wasatch SIAM Student Chapters Conference**

April 2026

University of Utah

- Coordinated with and advertised to Utah State University, Brigham Young University, Utah Valley University, Weber State University, and University of Utah
- Organized speaker schedule including an undergraduate student, graduate students, postdoctoral fellows, and a faculty member

- **Coordinator, Graduate Peer Mentorship Program**

Fall 2025

University of Utah

- Paired first-year graduate students with graduate student mentors
- Encouraged regular check-ins addressing academics, relocation, and graduate-student well-being

- **Chair, Graduate Student Recruitment Committee**

Spring 2025

University of Utah

- Led a five-member graduate student committee coordinating communication and a campus visit weekend for 23 prospective PhD students
- Paired prospective students with current graduate students and developed individualized schedules for faculty meetings, academic events, and informal community activities

HONORS AND AWARDS

- **RTG: Optimization and Inversion Grant**

Summer 2023 & Fall 2026

University of Utah Mathematics Department

- Research training grant provided to 1-2 graduate students per semester

- **Outstanding Graduate Student**

May 2025

University of Utah Mathematics Department

- Award provided to one graduate student annually based on research, teaching, and service

- **Fulbright Canada Mitacs Globalink Research Internship**

April 2021

Mitacs Globalink

- Awarded to U.S. undergraduate students to undertake advanced research projects in Canada

RESEARCH SOFTWARE

- **[CaratheodoryPruning.jl](#)**

Julia

Creator and maintainer

- Stable, scalable Carathéodory–Steinitz pruning algorithms for constructing positive quadrature rules

- **[ModelOrderReductionToolkit.jl](#)**

Julia

Creator and maintainer

- Projection-based model order reduction methods for parametric dynamical systems

INDUSTRY EXPERIENCE

- **Allianz Life** Summers 2021 and 2026
Hedging Intern Golden Valley, MN
 - Studied theory and implementation of valuation of financial derivatives and hedging
 - Implemented procedure for automating the assessment of incoming market data
 - Extended the hedge reporting software
 - Winapp development of tool to streamline software development process

CONFERENCES

- **SIAM Mathematics of Data Science (upcoming)** November 2026
Salt Lake City, UT
 - Talk: Greedy Frequency Domain Model Reduction for LTI Systems
 - Co-organized minisymposium: Dimension Reduction and Data Selection for Model Order Reduction
- **SIAM Imaging Science (upcoming)** November 2026
Salt Lake City, UT
 - Talk: Hemodynamic Monitoring with a Smartwatch
- **Wasatch SIAM Student Chapters Conference** April 2026
Salt Lake City, UT
 - Talk: Frequency Domain Greedy Model Order Reduction for LTI Systems
- **Joint Mathematics Meetings** January 2026
Washington, D.C.
 - Talk: Greedy Rational Approximation for Model Reduction of LTI Systems
- **Wasatch SIAM Student Chapters Conference** April 2025
Logan, UT
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **MAA North Central Section Meeting** March 2025
Northfield, MN
 - Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature
- **Model Reduction and Surrogate Modeling** September 2024
La Jolla, CA
 - Talk: Greedy Frequency Domain Model Reduction for Parametric Systems: New Theory and Algorithms
- **NSF Computational Mathematics PI Meeting** July 2024
Seattle, WA
 - Poster: Dynamic Bulk Conductivity in Radial Artery
- **Mathematical Opportunities in Digital Twins** December 2023
Arlington, VA
 - Poster: Dynamic Bulk Conductivity in Radial Artery